

Cursor Racer

1. Project Overview

A 2D Top Down racing game where you control the car by pointing the mouse cursor to the direction you want to go. There will be a single enemy AI you can race with, alongside multiple cars and originally made race tracks to choose from. (And potentially a vehicle tuning system if possible)

2. Project Review

This game is based on old-school 2D top down racing games. However, a new unique feature that is added is the ability to control the car entirely by using the mouse cursor.

3. Programming Development

3.1 Game Concept

The objective of this game is to win the race against an enemy AI, which will grant you credits to unlock new tracks and cars. The mechanic is simple, point your mouse cursor to where you want your car to go, and try to stay on track since going offroad will significantly slow you down.

3.2 Object-Oriented Programming Implementation

Car class :

Shared behavior for all cars (player and AI)

Main attributes :

x ,y , angle, speed, top_speed, turning_radius, off_road_speed (speed when the car is off road) etc.

Main methods :

update() runs every frame for movement and physics

check_waypoints() handles laps and checkpoints

draw() renders the car sprite

PlayerCar(Car) sub class :

Handles user controlled cars

Main attributes :

Inherited

Main methods :

control() main logic

mouse_steer() turn the car towards the cursor

auto_throttle() adjust speed based on the cursor distance

AICar(Car) sub class :

Handles AI controlled behavior

Main attributes :

Inherited

Main methods :

control() main logic

ai_steer() turns toward the next check point

ai_throttle() adjust speed based on turn

Track class :

Represent the racing track

Main attributes :

waypoints, width (track width), surface

Main methods :

build() generate the track graphics

on_track(x,y) check if a car is on the track

ParticleSystem class :

Handles visual effects

Main attributes :

particles, skids (skid marks)

Main methods :

build() generate the track graphics

on_track(x,y) check if a car is on the track

Game class :

Control the game loop and integrate everything

Main attributes :

Screen (display surface), clock (frame rate), track, race, hud

Main methods :

run() main loop

handle_events() : input handling

3.3 Algorithms Involved

The player and AI race on the same race track there are no collisions the AI follows an invisible path.

Player's car algorithm follows the cursor position.

To check if the car is off road or not we check if the car x y position is out of the track width or not.

4. Statistical Data (Prop Stats)

4.1 Data Features

	<i>Why is it good to have this data</i>	<i>How will you obtain 100 values of this data</i>	<i>Which variable or class will you collect this from</i>	<i>How will you display this feature data</i>
<i>Lap time</i>	<i>To show your performance and consistency</i>	<i>Record delta time and what lap the car is on every tick</i>	<i>Implement a time log inside update() and lap attribute to check the time and what lap the car is on</i>	<i>Table</i>
<i>Distance traveled</i>	<i>To show driving efficiency and acceleration</i>	<i>Record delta x and delta y each tick</i>	<i>Implement a previous position variable inside car class to find</i>	<i>Table</i>

	<i>patterns</i>		<i>the delta x,y</i>	
<i>Speed intervals</i>	<i>To analyze how twisty the track is and assist in choosing the right car</i>	<i>Record (timestamp, speed) every tick until reached 100 recorded data</i>	<i>In car class, speed is collected inside update()</i>	<i>Histogram</i>
<i>Time on-road - off-road</i>	<i>To show how many mistakes you make</i>	<i>Record weather the car is on road/ off road every tick until reached 100 recorded data</i>	<i>In track class using the on_track() method to check</i>	<i>Bar Chart</i>
<i>Speed-time</i>	<i>To analyze driving patterns (acceleration, braking, consistency)</i>	<i>Record (timestamp, speed) every tick until reached 100 recorded data</i>	<i>In car class, speed is collected inside update()</i>	<i>Line Graph</i>

4.2 Data Recording Method

The data will be recorded into a CSV file.

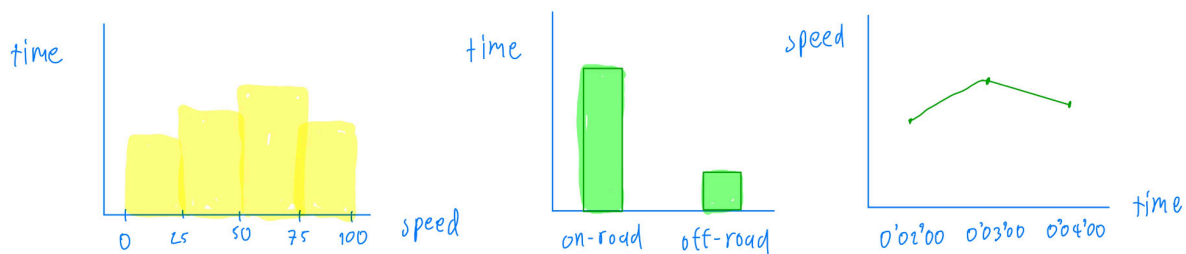
4.3 Data Analysis Report

Tools used : tkinter, pandas, matplotlib, seaborn

<i>Graph no.</i>	<i>Feature Name</i>	<i>Objective</i>	<i>Graph Type</i>	<i>X-axis</i>	<i>Y-axis</i>
<i>1.</i>	<i>Speed-Time</i>	<i>To show how the car's speed changes throughout the race</i>	<i>Line graph</i>	<i>Time</i>	<i>Speed</i>

2.	<i>Onroad-Offroad</i>	<i>Two bars to compare the time you spent on road and off road</i>	<i>Bar Graph</i>	<i>Onroad / Offroad</i>	<i>Time for each on road and off road</i>
3.	<i>Speed intervals</i>	<i>Histogram to tell which range of speed you spend the most time in</i>	<i>Histogram</i>	<i>Speed</i>	<i>Time</i>

Sketch



5. Project Planning Timeline

Week	Week	Task
1	8	Proposal submission / Project initiation
2	9	Make a working game prototype
3	10	Make different tracks and cars
4	11	Implement the GUI
5	12	Implement an economy system

6	13	Continue making the economy system (or car tuning system if there are time left)
7	14	Submission week (Draft)

6. Document version

Version: 1.0

Date: 12 March 2026

Editor: Phirawit Sommaichaiya

Version: 1.1

Date: 28 March 2026

Editor: Phirawit Sommaichaiya

Changes : Improved proposal according to TA suggestions

Version: 1.2

Date: 28 March 2026

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Changes : Added data tables